

AVANI J PATEL

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PROFESSIONAL SUMMARY

Software Engineer and Generative AI Developer with hands-on production experience building LLM-powered backend systems, RAG pipelines, and intelligent automation at Reliance Industries. Proficient in Python, FastAPI, LangChain, and LangGraph, with a strong foundation in vector search, prompt engineering, and scalable API design. Adept at architecting modular, production-grade AI systems end-to-end — from data ingestion through deployment.

EDUCATION

Bachelor of Technology, Computer Engineering

EXPECTED: MAY 2028

Birla Vishvakarma Mahavidyalaya (BVM Engineering College), Anand, Gujarat

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

AI Engineering Intern, Reliance Industries Limited (RIL)

MAY – JUN 2026

- Architected an enterprise AI Agent Configuration Platform to dynamically orchestrate LLM-powered workflows via Ollama (Llama 3, Qwen), enabling flexible, reusable multi-agent automation.
- Engineered a production Python backend with FastAPI, optimizing API response latency to support high-throughput enterprise requests.
- Designed an AI-powered KPI report generation agent using advanced prompt engineering and Jinja2 templating, significantly reducing manual reporting effort.
- Automated end-to-end report scheduling and delivery pipelines using APScheduler and SMTP, eliminating manual intervention.
- Integrated a React.js frontend with REST APIs and designed a modular backend architecture to support future PostgreSQL integration.

PROJECTS

Self-Healing RAG Pipeline | *Python, LangGraph, LangChain, FastAPI, Llama 3, FAISS, Hugging Face Embeddings*

- Architected a production-grade self-healing Retrieval-Augmented Generation (RAG) system using LangGraph stateful multi-agent orchestration for robust, hallucination-resistant question answering.
- Engineered an evaluation agent that verified response grounding against retrieved documents and automatically reformulated low-confidence queries for re-retrieval, improving answer reliability end-to-end.
- Built a modular pipeline with decoupled components for document ingestion, FAISS semantic search, LLM generation (Llama 3), evaluation, and graceful fallback — designed for future production deployment.

MarketPulse AI — Financial Intelligence Platform | *Python, RAG, FAISS, Llama 3, React, Pandas, NumPy, Matplotlib*

- Built an AI-powered financial analytics platform using RAG over a FAISS vector store to deliver context-aware, LLM-generated insights from earnings reports and financial news.
- Implemented quantitative technical indicators (RSI, MACD, Bollinger Bands, Moving Averages) alongside AI-generated analysis in interactive Streamlit dashboards for stock trend and volatility assessment.
- Optimized the document embedding and semantic retrieval pipeline for fast, accurate extraction of information from large unstructured financial corpora.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C#, JavaScript

AI / ML: Generative AI, LLMs (Llama 3, Qwen), Prompt Engineering, RAG, LangChain, LangGraph, Machine Learning, Neural Networks, Hugging Face

Web Development: React.js, FastAPI, Streamlit, REST APIs, Jinja2

Databases & Vector Search: FAISS, ChromaDB, PostgreSQL

Cloud & Tools: Microsoft Azure, Git, GitHub, Linux, VS Code

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS & LEADERSHIP

- NPTEL Elite Gold Certification — Business Fundamentals for Entrepreneurs
- Graphic Design Lead**, Computer Society of India (CSI), BVM Chapter
- Solved **100+ LeetCode problems** covering Data Structures & Algorithms.